

Global Minimum EQS Requirements for Substation Projects

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Process Area	Quality Management
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Target Group

This Instruction applies to the entire project organization, including all functions involved in project delivery such as Project Management, Engineering, Procurement, Contract Management, Site Execution, EQS, and all contractors and subcontractors in SE GT Substations.

Project and Site Management are responsible for implementing the requirements defined in this Instruction, while the EQS function provides governance, guidance, and compliance assurance.

Management Summary

This Instruction defines the minimum mandatory EQS baseline requirements for Substation project sites, covering Environment, Health & Safety (EHS), Quality Management (QM), and embedded Product Safety across the full project lifecycle, complemented by additional requirements scaled based on project risk, complexity, and category.

It establishes a consistent and structured approach to safe project execution, quality assurance, contractor management, and performance monitoring. In addition, it introduces a clear governance model for EQS, including compliance verification through audits and inspections, as well as escalation mechanisms for non-compliance.

The Instruction ensures alignment with Siemens Energy frameworks and international standards, including ISO 9001, ISO 14001, ISO 45001, and ISO 27001. It defines a globally consistent and auditable baseline while allowing controlled regional adaptation based on legal, contractual, and project-specific requirements.

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1. Purpose & Introduction

The purpose of this Instruction is to establish a standardized and enforceable EQS framework for Substation project execution.

EQS requirements shall be applied across the full project lifecycle, including Sales and Tendering phases, and shall be considered during project planning and supplier engagement. It defines the minimum requirements that must be in place to ensure safe execution, quality delivery, and the protection of people, environment, systems, and assets throughout all project phases, including mobilization, installation, commissioning, and handover.

The requirements defined in this Instruction are implemented within the Siemens Energy management system framework, which provides the structure for policies, processes, project lifecycle governance, and documentation. This ensures that EQS requirements are consistently applied, integrated into project execution, and supported by defined procedures, templates, and records.

The Instruction follows a risk-based and performance-oriented approach. Project organizations are accountable for execution, while the EQS function defines the framework, ensures alignment with company standards, and verifies compliance through structured governance mechanisms.

Product Safety is governed within the EQS framework; however, operational ownership shall be clearly defined between Engineering, Project Management and EQS at project level, in alignment with [SE Product Safety Standard](#).

This includes both technical aspects related to design, installation, and commissioning, as well as organizational aspects such as defined responsibilities, communication channels, and response mechanisms for safety-related issues.

2. Principles & Requirements

Safe, secure, and high-quality project execution relies on clear requirements, defined accountability, and consistent application across all project sites.

The requirements established in this Instruction represent the minimum mandatory baseline for EQS within Substation projects. These requirements define a mandatory baseline applicable to all projects and shall be complemented by additional requirements, governance activities, documentation, resources and KPIs shall be applied proportionately based on project category, complexity, contractual requirements, execution model and risk profile. Where legal, regulatory, contractual, or Siemens Energy requirements differ, the most stringent requirement shall apply.

Requirements are classified according to their level of obligation. Mandatory requirements are auditable and must be implemented without exception. Governance requirements define how compliance is ensured and monitored. Recommended practices support further improvement and maturity.

Security is governed through a dedicated Site Security framework and is not part of this Instruction.

Project Management is accountable for implementing the requirements defined in this Instruction. SE GT SU EO EQS owns the Instruction, governance framework and compliance verification process. Appointed EQS roles support implementation within their assigned areas of responsibility.

Implementation of this Instruction is supported by structured management system frameworks, which define how requirements are applied across the project lifecycle. These frameworks ensure alignment with defined project phases, structured deliverables, and clearly assigned process ownership.

The EQS operating model is based on a clear separation of responsibilities. Project organizations are responsible for planning and implementing EQS measures. The EQS function defines standards,

supports implementation, and verifies compliance through audits, inspections, performance monitoring, and escalation.

3. EQS Requirements for Project Sites

Projects shall implement the following minimum EQS requirements as part of project lifecycle activities, including Sales, Tendering, Engineering, Procurement, and Execution. These requirements are mandatory and subject to verification through governance processes. In consortium or partner-based execution models, contractual applicability shall be clarified, and alternative compliance mechanisms shall be defined where full contractual flow-down is not possible.

EQS requirements must be integrated into the overall project execution framework and aligned with project planning, execution activities, and key verification points. Project teams shall ensure that all EQS requirements are reflected in project processes, deliverables, and controls.

Minimum competencies, responsibilities and empowerment criteria for EQS roles shall be defined through applicable appointment letters and organizational governance requirements.

3.1. EHS Requirements

Projects shall establish and maintain a structured Environment, Health and Safety management approach prior to mobilization. Responsibilities must be clearly defined, and a competent EHS Manager in Projects (EHSMiP) must be formally appointed. EHS roles shall be defined based on project category and complexity and may include EHSMiP, Site EHS Manager, and additional site EHS resources.

Minimum qualification criteria and appointment timing for EHS roles shall be defined according to project milestones. A Project EHS Plan must be developed and approved at defined project milestones (e.g., prior to mobilization and aligned with PM phases). The Project EHS Plan shall be reviewed and approved by the Project Manager with support from the appointed EHSMiP.

Activities involving significant risk shall be supported by documented risk assessments and, where required, method statements prior to commencement of work. These assessments must include operational hazards as well as product and system-related risks, such as electrical hazards, energization activities, and system interfaces.

Operational controls must ensure safe execution of all activities. This includes implementation of permit-to-work systems, lock-out/tag-out procedures, and controls for high-risk activities. Particular attention must be given to energization and commissioning activities to ensure system integrity and prevent unintended release of energy.

Personnel, including contractors and subcontractors, must be adequately trained and competent for their assigned activities. This includes awareness of system-related risks associated with Substation equipment and installations.

Incidents, near misses, unsafe conditions, and unsafe actions must be reported, investigated, and addressed. High-potential events must be escalated to ensure appropriate visibility and management action. Investigation depth and management actions shall be proportionate to the actual and potential severity of the event.

Monitoring shall be ensured through inspections and site engagements. Deviations must be documented in Enablon and resolved in a timely manner. Environmental aspects and impacts shall be explicitly identified, assessed, and monitored.

3.2. Quality Management Requirements

Quality management shall be implemented through a structured and risk-based approach across all project phases.

The Project Quality Plan shall be defined based on project category, contractual requirements, and internal baseline expectations.

This plan defines quality objectives, applicable requirements, control processes, and responsibilities, including the integration of product safety aspects.

Quality in projects is supported by defined roles, including QMiPs and site-level QA/QC resources, where required, who ensure that quality processes are implemented, monitored, and continuously improved. Quality requirements shall align with existing Procurement and Supplier Quality Management processes. Responsibilities, authority, escalation rights and interfaces with Project Management, Engineering, Procurement and SQM shall be clearly defined.

Design and engineering activities must ensure that all technical and safety requirements are fulfilled. Product Safety shall be systematically integrated into design and engineering processes, ensuring that systems are safe to install, operate, and maintain. Open design issues, engineering changes and safety-critical deviations shall be formally assessed and approved before implementation or energization.

Inspection and testing must be performed in accordance with defined plans and requirements. Safety-critical elements must be verified to ensure system integrity and functional performance.

Non-conformance must be managed through a structured approach. All deviations from requirements must be documented, assessed, and resolved. Root cause analysis shall be performed for significant, recurring, high-potential or systemic issues, and corrective actions must be implemented and verified.

Supplier and contractor quality must be actively controlled. Qualification processes shall ensure capability, and contractual arrangements shall clearly define quality and safety expectations.

All project-related information must be controlled to ensure traceability, integrity, and confidentiality. Documentation must be protected against unauthorized access, loss, or manipulation. Access to critical data and systems must be managed in line with EQS security and information security requirements.

Information control shall be exercised through approved project documentation systems and defined access controls in accordance with Siemens Energy information management requirements.

Verification prior to energization shall be demonstrated through commissioning records, testing documentation, and client acceptance, where applicable.

Before handover, all systems must be verified. Systems shall not be energized without formal confirmation that all technical, quality, and safety requirements have been fulfilled.

Continuous improvement shall be ensured through the capture and application of lessons learned.

3.3. Non-Conformance Management

Projects shall establish and apply a structured approach to non-conformance management.

All deviations from defined requirements, including those related to safety, quality, or product integrity, must be documented and managed through approved project or organizational tools (e.g. Project Dash, JIRA or equivalent systems).

For significant or recurring issues, root cause analysis shall be performed to identify underlying causes. Corrective and preventive actions must be defined, implemented, and tracked to closure. The effectiveness of these actions must be verified. The selected tool shall be used consistently to ensure traceability and reporting visibility.

Non-conformance management must support continuous improvement by ensuring that lessons learned are captured and applied across projects.

3.4. Product Safety Requirements

Product Safety is governed within EQS, while operational responsibilities are assigned across Engineering, Project Management, and Procurement and must be ensured throughout the entire project lifecycle. Overall accountability resides with Project Management.

Projects must ensure that all products, systems, and installations comply with applicable technical standards and do not pose unacceptable risks to people, environment, or property.

Product Safety must be addressed through both technical and organizational measures. Technical aspects include design, engineering, installation, and commissioning controls. Organizational aspects include clear assignment of responsibilities, defined communication channels, and structured reporting mechanisms.

Substations require a system safety approach; system safety related risks must be identified and assessed with HAZID/HAZOP studies as part of project risk management. Appropriate control measures must be implemented, verified, and documented before systems are put into operation.

Product Safety requirements must also extend to instructions for transport, installation, commissioning, operation, maintenance, disassemble and disposal to ensure safe and intended use throughout the lifecycle.

Any product-related incidents or concerns must be reported, risk must be assessed, and managed through defined [Product Safety Issue Handling](#) processes. Where required, appropriate actions must be taken to protect employees, contractors, customers and third parties. Product Safety shall be included in internal audits using defined audit criteria.

3.5. Roles and Appointment Governance

Clear assignment of responsibilities is essential to ensure effective EQS implementation and auditability.

Key EQS roles must be formally appointed prior to mobilization, including responsibilities for EHS and Quality management. This includes roles such as EHS Managers in Projects and Quality Managers in Projects and Product Safety Responsible Person (where required).

Appointment letters shall define responsibilities, authority, accountability, escalation rights and stop-work authority (where applicable). They must be signed by the manager responsible and accepted by the appointed individual.

All appointment letters must be distributed to relevant stakeholders and stored within a controlled document system. These documents form part of the official project records and provide a clear audit trail.

No project shall commence without formally documented EQS role assignments. The EQS roles along with all the other project roles are documented in Project Organizational charts and their responsibilities defined accordingly.

3.6. EQS Governance and Enforcement

The EQS function ensures the consistent application of this Instruction through structured governance.

Compliance is verified through audits, inspections, performance monitoring, and management review processes. These mechanisms provide transparency and enable early identification of risks and deviations.

Non-compliance is addressed through a structured escalation approach. Minor deviations are corrected within the project. Severity levels (minor, significant, critical) shall be defined with clear escalation thresholds require formal action plans and escalation to management. In cases of critical risk, EQS has the authority to escalate and recommend suspension of work.

Supporting processes, templates, guidance documents and role descriptions shall be communicated and made available prior to full implementation of this Instruction to ensure consistent application across projects and regions.

These governance mechanisms ensure continuous improvement and effective implementation of EQS requirements across all project sites.

Compliance with this Instruction is mandatory for all Substation projects and forms part of project execution requirements. Non-compliance shall be treated as a deviation requiring formal corrective action.

4. Abbreviations, Definitions and Roles

4.1. Abbreviations

EQS – Environment, Health, Safety, Quality, and Security

EHS – Environment, Health & Safety

QM – Quality Management

QMIP – Quality Manager in Projects

EHSMIP – EHS Manager in Projects

NCR – Non-Conformance Report

CAPA– Corrective Action/Preventive Action

RCA – Root Cause Analysis

PTW– Permit to Work

Document Index

Standards and Instructions

No.	Title
000040	SE Standard Product Safety
007439	Product Safety Issue Handling@SE GT
005375	Product Safety Event Risk Assessment
008182	Quality Management in Project (QMiP) in SE GT
003619	General EHS Requirements for Contractors
	All SE EHS Standards and Instructions

References

Name/Title
NCC Overview - Sharepoint
SE GT EQS - Sharepoint
SE EQS SharePoint

Change History

Version	Date	Remarks
	Publication Date +3 years	Next Review
1	According to SEPH ("Publication Date")	First Publication

Organizational Terms

For abbreviations and definitions please refer to Siemens Energy's [organizational terms](#).

5. Annex

5.1. Application and Ownership of Annexes

The annexes defined in this Instruction form an integral part of the EQS framework and shall be considered mandatory components of this Instruction. They are not standalone documents but are to be applied as a combined and consistent package supporting project execution.

All annexes are assigned under the responsibility of the **Project Manager**, who is accountable for ensuring full implementation of this Instruction throughout the project lifecycle.

The Project Manager shall ensure that all annexes are acknowledged, applied, and maintained as part of the project's controlled documentation.

Project EQS roles, including EHSMIP, QMiP and additional site EQS roles where required are responsible for operational implementation and maintenance.

The EQS function retains governance responsibility, defining the framework, supporting implementation, and verifying compliance through audits and performance monitoring.

This ensures that requirements are clearly owned at project level, auditable, and consistently applied.

5.2. Annex – EQS Implementation and Governance Pack

The following annexes form an integral part of this Instruction and shall be implemented collectively to ensure effective EQS governance, execution, and verification.

5.2.1. Annex 1 - Project Manager Acknowledgement

Where applicable, this acknowledgement may be incorporated into the PM/CPM Letter of Empowerment (LoE) or equivalent project governance documentation. This acknowledgement applies to Project Managers and, where applicable, Site Managers and Field Supervisors responsible for project execution.

Project Name: [Insert Project Name]

Location: [Insert Location]

Project Number: [Insert Project Number]

Date: [DD/MM/YYYY]

Acknowledgement Statement

I hereby confirm that I have reviewed and understood the requirements defined in the **GT Substations Instruction - Minimum EQS Requirements for Project Sites**, including all associated annexes.

I acknowledge that these requirements define the **minimum mandatory EQS baseline** to be implemented throughout the project lifecycle.

Commitment

I confirm that:

The EQS requirements covering Environment, Health, Safety & Security, Quality, and Product Safety are integrated into project execution and planning. All required EQS roles are formally assigned, and responsibilities are clearly defined and accepted.

The required plans, including the Project EHS Plan and Project Quality Plan, are established, implemented, and maintained. Risk management, inspections, and verification activities are conducted in accordance with the Instruction.

Contractors and subcontractors are managed in line with defined EQS requirements. Non-conformances, incidents, and product safety concerns are properly reported, managed, and tracked to closure.

All relevant documentation, records, and evidence are maintained as part of the project's controlled documentation and are available for audit and verification.

Governance Awareness

I am aware that compliance with this Instruction will be verified through audits, inspections, and performance monitoring. Deviations will require corrective action and may result in escalation.

Declaration

I confirm my responsibility for implementing and maintaining EQS requirements throughout the project lifecycle.

Project Manager

Name: _____

Signature: _____

Date: _____

5.2.2. Annex 2 - Appointment Letter – EHSMIP

Project Name: [Insert Project Name]

Location: [Insert Location]

Effective Date: [DD/MM/YYYY]

Dear [Name],

You are hereby appointed as **EHS Manager in Projects (EHSMIP)** for the above-mentioned project.

You are responsible for ensuring the implementation and continuous improvement of Environment, Health, Safety & Security requirements throughout the project lifecycle. Responsibilities shall be executed in alignment with the Siemens Energy Zero Harm Framework and applicable EHS governance requirements.

You shall ensure that EHS planning is in place prior to mobilization, risks are identified and controlled, and all activities are executed safely. You are responsible for ensuring that incidents and near misses are reported, investigated, and closed, and that inspections and monitoring activities are carried out.

You shall ensure that contractors are integrated into the EHS framework and comply with project requirements. You are required to maintain all relevant documentation and records for audit purposes.

You are authorized to stop work in case of unsafe conditions and to escalate critical risks to project management and EQS.

Assigned by (Project Manager)

Name: _____

Signature: _____

Assigned by (EQS Organization)

Name: _____

Signature: _____

Accepted by (EHSMIP)

Name: _____

Signature: _____

5.2.3. Annex 3 - Appointment Letter – QMiP

Project Name: [Insert Project Name]

Location: [Insert Location]

Effective Date: [DD/MM/YYYY]

Dear [Name],

You are hereby appointed as **Quality Manager in Projects (QMiP)** for the above-mentioned project.

You are responsible for ensuring that all quality and product safety requirements are implemented, monitored, and continuously improved throughout the project lifecycle.

You shall ensure that a Project Quality Plan is established and applied, that inspection and testing activities are performed, and that all quality requirements are fulfilled. You are responsible for managing non-conformances, ensuring root cause analysis is performed where required, and that corrective actions are implemented and verified.

You shall ensure that supplier and contractor quality requirements are controlled and that all relevant documentation is maintained to ensure traceability and compliance.

You are authorized to escalate quality risks and require corrective actions for deviations from defined requirements, including escalation of critical risks to Project Management and EQS.

Assigned by (Project Manager)

Name: _____

Signature: _____

Assigned by (EQS Organization)

Name: _____

Signature: _____

Accepted by (QMiP)

Name: _____

Signature: _____

5.2.4. Annex 4 - EQS Audit Checklist

Objective

This Annex serves as a supplementary EQS verification checklist. Existing PM Milestone Gate Checklists or equivalent governance mechanisms may be used as alternative verification mechanisms where they cover the applicable EQS requirements.

1. EHS

Requirement	Status (✓/X)	Evidence	Remarks
EHSMP appointed		Appointment letter	
EHS Plan approved		Document	
Risk assessments completed		Records	
High-risk activities controlled		PTW / procedures	
Incidents reported and tracked		Logs/ Enablon	
Inspections conducted		Records	

2. Quality

Requirement	Status	Evidence	Remarks
QMIP appointed		Appointment letter	
Quality Plan available		PQP	
Inspection & test activities defined		ITPs	
Quality records maintained		Documentation	
Supplier quality controlled		Approvals / audits	

3. Non-Conformance

Requirement	Status	Evidence	Remarks
NCR system implemented		System records	
NCRs tracked to closure		Logs	
Root cause analysis performed		Reports	
Corrective actions verified		CAPA records	

4. Product Safety

Requirement	Status	Evidence	Remarks
Product safety risks assessed		HAZID/HAZOP Risk register	
Safety integrated in inspections		ITPs	
Verification before energization		Records	
Incidents reported		Logs	

5.2.5. Annex 5 - KPI Framework

The KPI framework defines a scalable set of indicators that may be tailored according to project category, complexity, risk profile, scope and contractual requirements. Not all KPIs will be applicable to every project.

KPI definitions, data sources, reporting frequencies and accountability shall be maintained within supporting guidance documentation to ensure consistency and comparability across projects.

A balanced KPI set shall be maintained, covering both **leading indicators (prevention-focused)** and **lagging indicators (outcome-based)**.

5.2.5.1. EHS KPIs

Lagging Indicators (Outcome-Based)

- Total Recordable Incidents (TRI)
- Total Recordable Incident Rate (TRIR)
- High Potential Incidents (HiPo)
- First Aid Cases
- Environmental incidents
- Property damage incidents

Leading Indicators (Prevention-Focused/ Proactive Reports & rate)

- Number of risk workshops conducted
- Number of unsafe conditions / hazard observations reported
- Number of Near Misses.
- Number of safety positive observations / behavioral observations
- Leadership engagements (site visits, safety walks, Zero Harm engagements)
- Training completion rate (mandatory training / certifications)
- Number of Permit-to-Work activities.
- Number and closure rate of corrective actions from inspections and audits
- Number and closure rate of lessons learned and risk assessments

5.2.5.2. Quality KPIs

Leading Indicators

- Number of inspections
- Internal and external audit completion rate
- Supplier qualification and audit performance

Lagging Indicators

- Number of Non-Conformance Reports (NCRs)
- Non-Conformance Cost (NCC as % of project value or cost baseline)
- Customer complaints / deviations

5.2.5.3. Non-Conformance & CAPA KPIs

NCR Performance

- Total number of NCRs raised
- NCR trend (increase / decrease over time)
- NCR closure rate (%)
- Number of overdue NCRs

CAPA Performance

- Number of CAPAs open
- Number of overdue CAPAs
- CAPA closure rate (%)
- Average time to closure
- Recurrence of similar issues (repeat findings)

Root Cause Analysis (RCA)

- Number of RCAs initiated
- Number of RCAs completed
- Number of overdue RCAs

Percentage of significant issues with completed RCA

5.2.5.4. Product Safety KPIs

(Product Safety KPI application shall align with project-specific Product Safety responsibilities and may include Safety by Design (SbD) assessments where applicable.)

Leading Indicators

- Product safety reviews completed (design / installation / commissioning)
- Safety checks integrated into inspection and testing activities
- Product safety risk assessments completed

- Verification activities prior to energization

Lagging Indicators

- Product safety incidents (including property damage or safety-critical events)
- Product-related non-conformances
- Safety-critical deviations identified at commissioning

5.2.5.5. Governance KPIs

(Product Safety verification shall be incorporated into existing Quality/EHS/QHSE audit activities where applicable.)

- Number of audits performed (internal / external)
- Number of audit findings
- Audit closure rate (%)
- Number of overdue audit actions
- KPI reporting compliance (on-time reporting)
- Management review completion

5.2.5.6. KPI Application and Review

KPIs shall be:

- Defined and applied at project level based on project risk, scope, and complexity
- Measured using consistent and reliable data sources
- Reviewed regularly by Project Management and EQS
- Used to identify trends, risks, and improvement opportunities
- Linked to corrective and preventive actions where required

A clear distinction shall be maintained between:

- **Leading indicators**, which provide early warning of potential risks and measure preventive actions
- **Lagging indicators**, which reflect actual performance and events that have occurred

Both types of indicators shall be used in combination to ensure effective performance management and risk control.

5.2.5.7. Escalation

Where KPI trends indicate increased risk, recurring issues, or non-compliance:

- Corrective actions shall be initiated
- Issues shall be escalated through the defined governance structure
- Additional controls or measures shall be implemented as required